# Calcurion Maths Worksheets

A-Level Further Mathematics

**Topic:** Further Mechanics 1 – Circular Motion

Medium & Hard • 25 Questions

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## Section A – Medium (Q1–15)

**1.** A particle moves in a horizontal circle of radius 2 m at constant speed $4\,\mathrm{m\,s^{-1}}$. Find the centripetal acceleration. **[2]**

**2.** Find the angular speed $\omega$ if the period is 2 s. **[2]**

**3.** A particle of mass 0.5 kg moves in a horizontal circle of radius 1 m at $3\,\mathrm{m\,s^{-1}}$. Find the centripetal force. **[3]**

**4.** State the relationship $v = r\omega$. **[1]**

**5.** A car of mass 1000 kg takes a flat bend of radius 50 m at $20\,\mathrm{m\,s^{-1}}$. Find the lateral friction required. **[3]**

**6.** A particle on a string of length 0.5 m moves in a horizontal circle with the string at $30^\circ$ to the vertical (conical pendulum). Find the speed. **[5]**

**7.** Find the tension in the string for the conical pendulum above if mass is 0.2 kg. **[3]**

**8.** A particle moves in a vertical circle of radius $r$ on the end of a string. Find the tension at the lowest point if speed there is $u$. **[3]**

**9.** Find the condition that the particle completes a vertical circle on a string. **[3]**

**10.** A particle slides on the inside of a smooth vertical circular track of radius $r$. Find the speed at angle $\theta$ from the lowest point if it starts from rest at the top. **[4]**

**11.** Find the reaction at the top of a vertical circle for a particle of mass $m$ with speed $v$. **[3]**

**12.** A car banks at angle $\alpha$ on a curve of radius $r$. Find the design speed with no lateral friction. **[4]**

**13.** Convert $60\,\text{rpm}$ to $\text{rad}\,\text{s}^{-1}$. **[2]**

**14.** A particle of mass $m$ is attached to a fixed point by a string of length $l$ and moves in a horizontal circle with the string making angle $\alpha$ with the vertical. Show that $\omega^2 = \frac{g}{l\cos\alpha}$. **[4]**

**15.** Find the period of the conical pendulum in terms of $l$ and $\alpha$. **[3]**

## Section B – Hard (Q16–25)

**16.** A particle is projected from the lowest point of a smooth vertical circular tube of radius $r$ with speed $\sqrt{5gr}$. Find where it leaves the circle. **[6]**

**17.** A car of mass $m$ negotiates a banked curve of angle $\alpha$ and radius $r$ at speed $v$. Find the lateral friction force if $\mu$ is limiting. **[6]**

**18.** A particle on a string moves in a vertical circle. At the highest point the tension is zero. Find the speed at the lowest point. **[4]**

**19.** A bead slides on a smooth vertical circular wire. It starts from rest at an angular position $\alpha$ from the highest point. Find its speed at the bottom. **[4]**

**20.** A particle attached to two strings of equal length moves in a horizontal circle. Set up equations for tensions. **[5]**

**21.** Find the maximum speed of a car on a flat curve of radius $r$ if $\mu = 0.5$. **[3]**

**22.** A conical pendulum has string length 1 m and period 1.5 s. Find the angle to the vertical. **[5]**

**23.** A particle moves in a vertical circle with centre $O$. At angle $\theta$ to the vertical the speed is $v$. Write $T - mg\cos\theta = \frac{mv^2}{r}$ or similar and energy equation. **[4]**

**24.** A satellite orbits Earth in a circular orbit of radius $R$. Find the period in terms of $g$ and $R$ (approx near surface). **[4]**

**25.** A particle is released from rest at the top of a smooth sphere of radius $r$. Find where it leaves the surface. **[6]**

*End of Worksheet* • *Show all working clearly. Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless otherwise stated.*